Diffusion-Based Object Detection in Aerial Images

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SELF-DECLARATION

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Abstract

The suitability of diffusion-based object detection for aerial imagery was examined through horizontal and oriented bounding-box formulations. DiffusionDet was fine-tuned and evaluated on DIOR, NWPU VHR-10, and HRSC2016. The mathematical formulation of diffusion models, iterative box denoising, Gaussian oriented-box representation, rotation equivariance, and feature alignment was studied. Oriented evaluation records were also examined for DIOR-R, HRSC2016, and DOTA-v1.0. All available checkpoint metrics were tabulated, and dataset-wise comparisons were made with published detectors using the closest matching box representation and reported metric. The strongest recorded HBB AP values were 45.4766% on DIOR, 61.76% on NWPU VHR-10, and 84.4688% on HRSC2016. The DIOR-R GSDet inference-only result yielded 46.23% mAP. The local HRSC2016 OBB result yielded 72.32% mAP but remained provisional because checkpoint and data-prefix inconsistencies had been recorded. The DOTA-v1.0 output reported 47.6231% COCO-style mAP, with the evaluation subset and protocol requiring further verification. HBB and OBB results were kept separate, and comparisons with the literature were qualified wherever the evaluation settings were not identical.

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1 Introduction

Aerial object detection was investigated as a localization and recognition problem in which objects had to be identified despite large changes in scale, orientation, density, and background. Diffusion-based detectors were examined as an alternative to conventional candidate-based localization.

1.1 Literature Survey

Conventional object detectors were commonly based on anchors, region proposals, or learnable queries. Although strong detection performance had been obtained with these approaches, the generation and assignment of candidate objects remained important design components. DiffusionDet introduced a denoising formulation in which noisy bounding boxes were iteratively refined towards object locations, and detection was treated as set prediction over progressively denoised boxes [1]. The formulation was considered relevant to aerial imagery because candidate geometry could be updated over several reverse steps rather than being predicted only once.

For aerial scenes containing elongated or rotated instances, horizontal boxes could include substantial background and could not represent object orientation explicitly. Oriented detection methods had therefore been developed to represent object geometry more accurately. OrientedDiffDet used an oriented box parameterization and feature alignment, while ReDiffDet represented oriented boxes with two-dimensional Gaussian statistics and introduced a rotation-equivariant diffusion process [3, 2]. These approaches were regarded as complementary extensions of diffusion-based localization.

The DIOR benchmark was introduced for object detection in optical remote sensing imagery, and NWPU VHR-10 was used as a benchmark for detecting objects across ten aerial categories [4, 5]. HRSC2016 was designed for ship detection, while DOTA-v1.0 provided oriented annotations for objects with arbitrary directions [6, 7]. For comparisons, the results reported by MSA-YOLO on DIOR, by the adaptive feature-learning detector on NWPU VHR-10, and by GSDet and OrientedDiffDet on the oriented benchmarks were used as literature references [8, 9, 10, 3].

1.2 Objectives and Work Carried Out

The theoretical formulation and denoising architecture of DiffusionDet were studied, and a working detection environment was configured. DiffusionDet was fine-tuned and evaluated on DIOR, NWPU VHR-10, and HRSC2016 using horizontal bounding boxes. The checkpoint-wise AP, AP50, AP75, APs, APm, and APl values available in the experiment records were collected and organized into tables. The Gaussian representation and rotation-equivariant formulation of ReDiffDet, together with the proposal and feature-alignment design of OrientedDiffDet, were studied. Oriented inference or evaluation outputs available for DIOR-R, HRSC2016, and DOTA-v1.0 were tabulated and compared with relevant published detectors. Differences in data splits, checkpoint status, and evaluation protocols were recorded so that results from distinct detection settings were not combined into a single ranking.

2 Diffusion Models and Their Variations

2.1 Probabilistic formulation of diffusion models

A diffusion model was considered as a latent-variable generative model in which a data distribution was mapped to a simple reference distribution through a sequence of stochastic transitions, while generation was performed by learning the reverse sequence. A clean sample was denoted by $\boldsymbol{z}_0 \sim q(\boldsymbol{z}_0)$ and its progressively corrupted states by $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_T$. The forward process was defined as the Markov chain $q(\boldsymbol{z}_{1:T} \mid \boldsymbol{z}_0) = \prod_{t=1}^{T} q(\boldsymbol{z}_t \mid \boldsymbol{z}_{t-1})$, so that, given the immediately preceding state, the next state was independent of all earlier states. For the variance-preserving Gaussian process, each transition was given by $q(\boldsymbol{z}_t \mid \boldsymbol{z}_{t-1}) = \mathcal{N}(\boldsymbol{z}_t; \sqrt{\alpha_t}\,\boldsymbol{z}_{t-1}, \beta_t \boldsymbol{I})$ with $\alpha_t = 1 - \beta_t$, where $\beta_t \in (0, 1)$ controlled the noise added at step t. A schedule of small noise increments was used so that the sample was transformed gradually rather than destroyed in a single step.

2.2 Closed-form forward noising

With the cumulative product $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, repeated substitution of the one-step transition gave $\boldsymbol{z}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{z}_0 + \sqrt{1 - \bar{\alpha}_t}\,\boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, or equivalently $q(\boldsymbol{z}_t \mid \boldsymbol{z}_0) = \mathcal{N}(\boldsymbol{z}_t; \sqrt{\bar{\alpha}_t}\,\boldsymbol{z}_0, (1 - \bar{\alpha}_t)\boldsymbol{I})$. The marginal at any timestep could therefore be sampled directly, and training did not require every transition from 1 to t to be simulated. At early timesteps $\bar{\alpha}_t$ was close to one and much of the original signal was retained, whereas at later timesteps the noise dominated; with a suitable schedule, $q(\boldsymbol{z}_T)$ approached $\mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$.

2.3 The reverse process and posterior distribution

The learned reverse process was modelled as $p_\theta(\boldsymbol{z}_{0:T}) = p(\boldsymbol{z}_T) \prod_{t=1}^{T} p_\theta(\boldsymbol{z}_{t-1} \mid \boldsymbol{z}_t)$ with $p(\boldsymbol{z}_T) = \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, and each reverse transition was parameterized as a Gaussian, $p_\theta(\boldsymbol{z}_{t-1} \mid \boldsymbol{z}_t) = \mathcal{N}(\boldsymbol{z}_{t-1}; \boldsymbol{\mu}_\theta(\boldsymbol{z}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{z}_t, t))$, whose mean was predicted by a neural network. For the forward Gaussian chain, the posterior conditioned on the clean sample was also Gaussian, $q(\boldsymbol{z}_{t-1} \mid \boldsymbol{z}_t, \boldsymbol{z}_0) = \mathcal{N}(\boldsymbol{z}_{t-1}; \tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t \boldsymbol{I})$, with $\tilde{\beta}_t = \dfrac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$ and $\tilde{\boldsymbol{\mu}}_t = \dfrac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t\,\boldsymbol{z}_0 + \sqrt{\alpha_t}\,(1 - \bar{\alpha}_{t-1})\,\boldsymbol{z}_t}{1 - \bar{\alpha}_t}$. This result followed from Bayes' rule applied to the likelihood $q(\boldsymbol{z}_t \mid \boldsymbol{z}_{t-1})$ and the prior $q(\boldsymbol{z}_{t-1} \mid \boldsymbol{z}_0)$, both of which were Gaussian in $\boldsymbol{z}_{t-1}$. The posterior provided the target reverse transition during training, whereas at inference $\boldsymbol{z}_0$ was unknown and was therefore estimated by the network, either directly or through the noise component, from $\boldsymbol{z}_t$.

2.4 Training objectives: likelihood and denoising

Diffusion probabilistic models were trained by optimizing a variational bound on the data likelihood, $-\log p_\theta(\boldsymbol{z}_0) \le L_T + \sum_{t=2}^{T} L_{t-1} + L_0$, in which $L_T = D_{\mathrm{KL}}(q(\boldsymbol{z}_T \mid \boldsymbol{z}_0) \,\|\, p(\boldsymbol{z}_T))$ was the terminal prior-matching term, $L_{t-1} = \mathbb{E}_{q(\boldsymbol{z}_t \mid \boldsymbol{z}_0)}[D_{\mathrm{KL}}(q(\boldsymbol{z}_{t-1} \mid \boldsymbol{z}_t, \boldsymbol{z}_0) \,\|\, p_\theta(\boldsymbol{z}_{t-1} \mid \boldsymbol{z}_t))]$ were the reverse-transition terms, and $L_0 = -\mathbb{E}_{q(\boldsymbol{z}_1 \mid \boldsymbol{z}_0)}[\log p_\theta(\boldsymbol{z}_0 \mid \boldsymbol{z}_1)]$ was the reconstruction term. The KL terms encouraged the learned reverse transitions to match the true forward posterior. A widely used parameterization predicted the noise $\boldsymbol{\epsilon}$, which led to the simplified denoising objective $\mathcal{L}_{\mathrm{noise}} = \mathbb{E}_{\boldsymbol{z}_0, t, \boldsymbol{\epsilon}}\big[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{z}_t, t)\|_2^2\big]$, and an equivalent parameterization predicted $\boldsymbol{z}_0$ directly. These objectives described the general learning principle of diffusion models. An object detector, however, had additionally to assign predictions to object instances and classify them, and the detection methods discussed below therefore used detection-specific losses rather than treating detection as unconditional image generation.

2.5 Sampling and the role of the noise schedule

At inference, a sample was initialized as $\boldsymbol{z}_T \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ and refined repeatedly through $\boldsymbol{z}_T \to \boldsymbol{z}_{T-1} \to \cdots \to \boldsymbol{z}_0$. The noise schedule determined the signal-to-noise ratio at each timestep and hence the difficulty of the denoising task: a schedule that removed useful signal too quickly made recovery difficult, whereas an overly conservative schedule required more steps. DDIM-style sampling used a subsequence of timesteps and reduced the number of network evaluations, and DPM-Solver solved the corresponding diffusion ordinary differential equation with fewer function evaluations. Such accelerated samplers were particularly relevant to detection, where each network evaluation involved feature extraction for many candidate boxes.

2.6 DiffusionDet: applying diffusion to bounding boxes

DiffusionDet adapted this generative formulation to object detection [1]. Instead of diffusing image pixels, the diffusion state was a set of candidate boxes $\boldsymbol{b}_0 \in \mathbb{R}^{N \times 4}$ with $\boldsymbol{b}_i = (c_{x,i}, c_{y,i}, w_i, h_i)$. The forward process corrupted the ground-truth boxes. Since the number of ground-truth instances varied from image to image

whereas the detector operated on a fixed number of box slots, the ground-truth set was padded to a fixed training cardinality before corruption, and concatenating random boxes was reported as an effective padding strategy. The box coordinates were scaled consistently because the signal-to-noise ratio had a substantial effect on detection performance.

The image encoder extracted multi-scale features with a backbone and a feature pyramid network (FPN). The corrupted boxes were used to sample region-of-interest features from these maps, and a cascaded detection decoder predicted class scores and refined box coordinates. The model was conditioned on the image and the timestep, $(\hat{\boldsymbol{b}}_0, \hat{\boldsymbol{c}}) = f_\theta(\boldsymbol{b}_t, t, \boldsymbol{x})$, where $\hat{\boldsymbol{c}}$ denoted the predicted class scores. In contrast to fixed-query detectors, the candidate boxes were not learned static queries; they were noisy box states that changed during the reverse process.

The training objective combined set prediction with classification and localization supervision. Multiple predictions could be assigned to one ground-truth instance using a top-k matching strategy based on prediction cost and optimal-transport assignment, so that the objective supervised the predicted object set rather than only a generic noise-prediction loss. The decoder parameters were shared across the denoising stages, with timestep embeddings informing the network of the current noise level. At inference, boxes were sampled from a Gaussian distribution and cleaned iteratively by the decoder, and box renewal could be used to replace low-quality candidates. The number of proposals and denoising steps could be varied at evaluation time, which created a trade-off between computational cost and accuracy. This iterative refinement was the central difference between DiffusionDet and detectors that made a single forward prediction from fixed learned queries.

2.7 Oriented detection and Gaussian box representation

An oriented bounding box was represented by its centre, width, height, and angle, $b = (c_x, c_y, w, h, \theta)$. ReDiffDet represented this rectangle by a two-dimensional Gaussian $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (c_x, c_y)^\mathsf{T}$ and $\boldsymbol{\Sigma} = \boldsymbol{R}\boldsymbol{\Lambda}\boldsymbol{R}^\mathsf{T}$, where $\boldsymbol{R} = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ and $\boldsymbol{\Lambda} = \operatorname{diag}\big((w/m)^2, (h/m)^2\big)$ [2]. Here m was a scaling constant used in the parameterization of the paper. The eigenvectors of the covariance encoded the principal directions of the rectangle and the eigenvalues encoded its dimensions, so that orientation was embedded in the covariance matrix rather than treated as an isolated scalar, and a rotation of the box corresponded to a consistent rotation of the mean and covariance.

ReDiffDet defined a diffusion process over these Gaussian box states. The theoretical motivation was rotation equivariance: a rotation of the input geometry should rotate the predicted geometry in the same way. The terminal isotropic Gaussian was invariant to rotations, which provided a suitable starting distribution for reverse sampling. A conditional encoder was further introduced to generate spatially meaningful conditional boxes and to reduce uncontrolled random placement, and an aligned decoder combined rotation-aware region feature extraction (RiRoI Align), self-attention, and instance interaction to align candidate boxes with image features [2].

2.8 OrientedDiffDet and feature alignment

OrientedDiffDet used a six-parameter representation $(x, y, w, h, \Delta\alpha, \Delta\beta)$, in which (x, y) and (w, h) described the enclosing horizontal rectangle and the offsets described the oriented geometry. Oriented proposals were constructed from noisy box states, and the network learned to reverse the corruption process [3]. The architecture contained an image encoder with a backbone and FPN, followed by a detection decoder. The oriented proposal was used to extract region features, which were passed through feature alignment and classification and regression layers. The alignment operation converted the proposal geometry into a directional rectangular region so that the sampled features corresponded better to the orientation of the object, which was important for elongated aerial objects because pooling from an axis-aligned region could include unnecessary background. DPM-Solver was also adopted to accelerate the reverse sampling. OrientedDiffDet was included as a related formulation, and the numerical tables below were restricted to the runs recorded in the result files.

3 Methodology

3.1 Architecture of Diffusion-Based Object Detectors

3.1.1 DiffusionDet Architecture

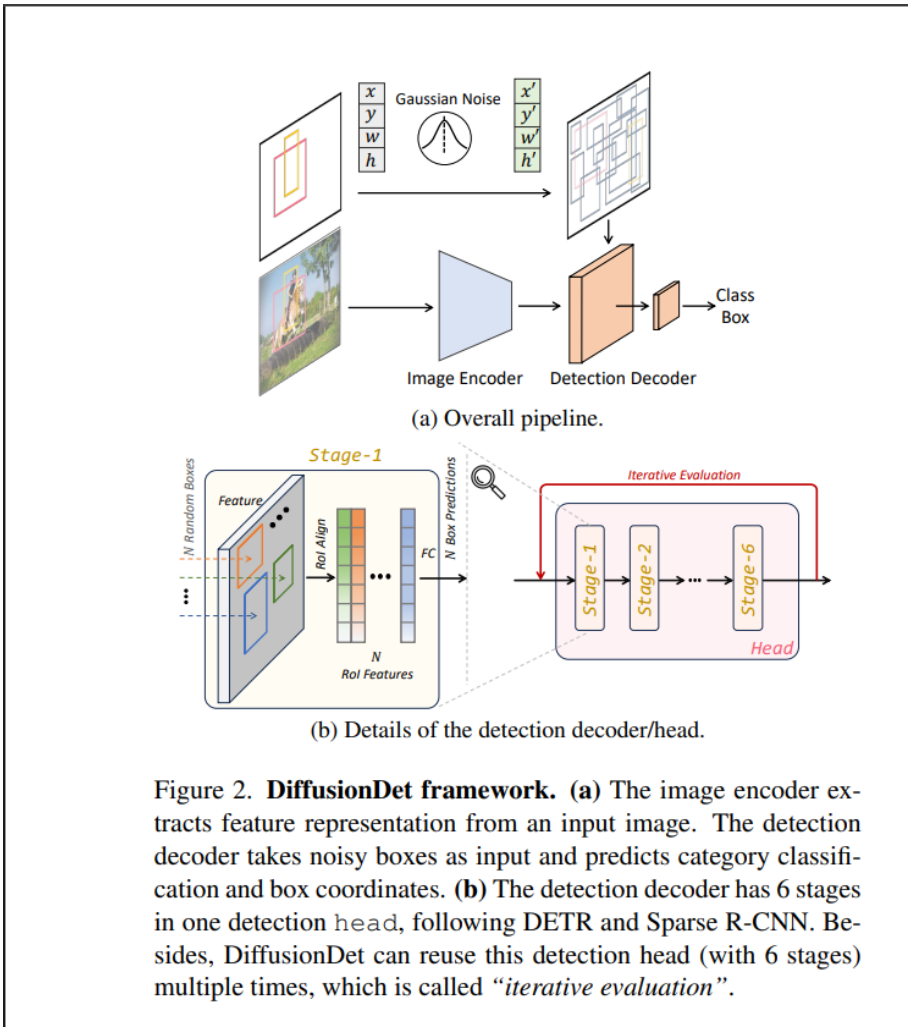


Figure 2. **DiffusionDet framework. (a)** The image encoder extracts feature representation from an input image. The detection decoder takes noisy boxes as input and predicts category classification and box coordinates. **(b)** The detection decoder has 6 stages in one detection `head`, following DETR and Sparse R-CNN. Besides, DiffusionDet can reuse this detection head (with 6 stages) multiple times, which is called *"iterative evaluation"*.

Figure 1: Overall architecture of DiffusionDet, showing the image encoder, noisy bounding-box proposals, RoI feature extraction, and multi-stage detection decoder.

DiffusionDet formulates object detection as a generative denoising process in which noisy bounding-box proposals are progressively refined to obtain accurate object locations and categories. Unlike conventional object detectors that directly generate bounding boxes from image features, DiffusionDet treats bounding boxes as the variables of a diffusion process. A detection network is trained to recover meaningful object locations from noisy bounding-box representations.

Image Encoding and Noisy Box Generation. As illustrated in Fig. 1, the input image is first processed by an image encoder to extract visual feature representations containing spatial and semantic information. In parallel, a set of bounding-box proposals is generated by sampling from a Gaussian distribution. Each proposal is represented using four parameters:

$$b_t = (x_t, y_t, w_t, h_t), \tag{1}$$

where (x_t, y_t) denotes the centre coordinates, while w_t and h_t represent the width and height of the bounding box.

During training, Gaussian noise is progressively introduced into ground-truth bounding boxes through the forward diffusion process. The noisy representation at timestep t can be expressed as

$$b_t = \sqrt{\bar{\alpha}_t}\, b_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{2}$$

where b_0 denotes the original bounding box, $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, $\alpha_t = 1 - \beta_t$, and β_t represents the noise variance schedule. This formulation allows noisy bounding boxes to be generated directly at arbitrary diffusion timesteps.

Detection Decoder and Iterative Refinement. The noisy bounding-box proposals are combined with the image features and supplied to the detection decoder. Region-specific features are extracted from the image feature maps using RoIAlign, which produces fixed-size feature representations for the candidate regions. These features are subsequently processed by the detection head to predict object categories and refined bounding-box coordinates.

The detection decoder consists of six stages within a single detection head and follows the design principles of DETR and Sparse R-CNN. At each stage, the candidate boxes are refined using the extracted image features. The decoder can be represented abstractly as

$$(p_i, b_i^{\text{pred}}) = \mathcal{D}_\theta(f_i, b_i^{\text{in}}), \tag{3}$$

where f_i denotes the feature representation of the i-th proposal, p_i represents its predicted class probabilities, and b_i^{in} and b_i^{pred} denote the input and refined bounding boxes, respectively.

The reverse diffusion process learns to transform noisy proposals into meaningful object detections. During inference, the detection head can also be reused multiple times, with the predictions from one evaluation serving as inputs to the next. This procedure is known as *iterative evaluation* and enables progressive refinement of the predicted bounding boxes. Thus, the combination of diffusion-based box generation and region-based feature processing provides a generative approach to object detection.

3.1.2 ReDiffDet Architecture

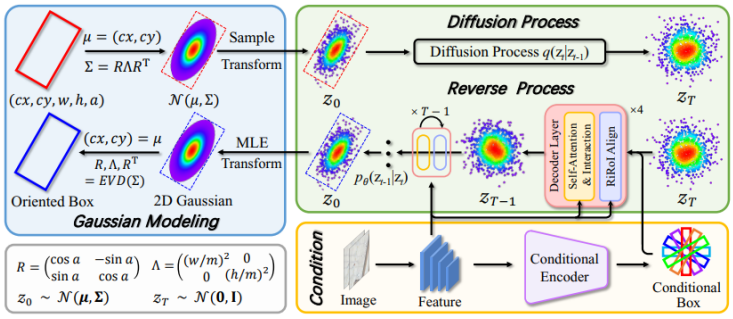


Figure 3. **Overview of ReDiffDet.** We represent an oriented box as a 2D Gaussian distribution. Gaussian distributions of boxes diffuse into standard normal distribution in the diffusion process. ReDiffDet generates boxes using the reverse process in inference. The conditional encoder provides conditional boxes. The aligned decoder layer consists of RiRoI Align, self-attention, and instance interaction.

Figure 2: Overview of ReDiffDet, illustrating Gaussian modelling of oriented bounding boxes, forward diffusion, conditional reverse denoising, and the aligned decoder architecture.

ReDiffDet extends the diffusion-based detection formulation to oriented object detection by representing oriented bounding boxes as two-dimensional Gaussian distributions. Instead of directly modelling the geometric parameters of an oriented box during diffusion, its position, dimensions, and orientation are encoded through the mean and covariance of a Gaussian distribution. This provides a probabilistic representation of object geometry that can be used in the diffusion and denoising processes.

Gaussian Representation of Oriented Bounding Boxes. An oriented bounding box is represented using five parameters:

$$b = (c_x, c_y, w, h, a), \tag{4}$$

where (c_x, c_y) denotes the centre coordinates, w and h represent the width and height, and a denotes the rotation angle.

The corresponding two-dimensional Gaussian distribution is represented as

$$\mathcal{N}(\mu, \Sigma), \tag{5}$$

where the mean vector specifies the centre of the box:

$$\mu = \begin{bmatrix} c_x \\ c_y \end{bmatrix}. \tag{6}$$

The covariance matrix encodes the dimensions and orientation of the box and is expressed as

$$\Sigma = R\Lambda R^T, \tag{7}$$

where

$$R = \begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} (w/2)^2 & 0 \\ 0 & (h/2)^2 \end{bmatrix}. \tag{8}$$

Here, R represents the rotation matrix, while Λ contains the squared half-dimensions of the box. Consequently, the covariance matrix captures both the spatial extent and orientation of the object. The Gaussian representation can subsequently be converted back into oriented bounding-box parameters through eigendecomposition, in which the eigenvectors and eigenvalues provide the principal directions and corresponding dimensions.

Forward and Reverse Diffusion Processes. As shown in Fig. 2, the forward diffusion process gradually transforms the Gaussian representations of ground-truth oriented boxes into noisy distributions. Let z_0 denote the initial Gaussian representation. The forward process is modelled as a Markov chain:

$$q(z_t \mid z_{t-1}), \tag{9}$$

where Gaussian perturbations are progressively introduced at successive timesteps. After a sufficiently large number of diffusion steps, the representation approaches a standard normal distribution:

$$z_T \sim \mathcal{N}(0, I). \tag{10}$$

The reverse diffusion process learns to recover the original oriented-box representations from the noisy distributions. It is conditioned on image information and can be represented by the reverse transition

$$p_\theta(z_{t-1} \mid z_t, I), \tag{11}$$

where I denotes the input image. The image is processed by a feature extraction network, and the resulting features are supplied to a conditional encoder. This encoder generates image-dependent box information that guides the reverse denoising process towards valid object locations and orientations. The process is repeated iteratively until the final Gaussian representations are obtained and converted into oriented bounding boxes.

Aligned Decoder Architecture. The decoder incorporates three principal components: RiRoIAlign, self-attention, and instance interaction. RiRoIAlign extracts region-specific features aligned with the orientations of candidate bounding boxes, allowing features to be obtained from arbitrarily oriented object regions. This is particularly useful for aerial images, where objects may appear at different rotation angles.

The extracted features are processed through self-attention, which captures relationships within the feature representations. Instance interaction further facilitates information exchange between different object proposals, enabling the detector to model relationships among candidate instances. These components operate within the conditional reverse diffusion framework and contribute to the progressive refinement of oriented bounding-box predictions.

Overall, ReDiffDet combines Gaussian modelling of oriented bounding boxes, forward diffusion, image-conditioned reverse denoising, and an aligned decoder. The Gaussian representation provides a unified way to model object position, dimensions, and orientation, while the conditional decoder uses image features to guide the generation of oriented detections.

3.2 Implementation Environment

The experiments were conducted in a Linux/WSL environment with Python-based deep-learning frameworks. The explicitly documented evaluation of all Datasets for ReDiffDet and DiffDET used Python 3.10, PyTorch 2.4.0+cu121, MMEngine 0.10.7, and an NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB VRAM. The same GPU family was recorded in other experiment logs. Exact software versions were not captured consistently for

every run, and the environment was therefore not assumed to be identical across all datasets.

3.3 Horizontal Bounding-Box Experiments

DiffusionDet was fine-tuned on three datasets: DIOR, NWPU VHR-10, and HRSC2016. For DIOR, the training and evaluation datasets were registered as `dior_trainval` and `dior_test`, respectively. The fine-tuning procedure comprised 11,725 iterations and required 1 hour, 17 minutes, and 32 seconds for each epoch as number of batches was 1. The resulting complete-epoch checkpoint was subsequently evaluated on the DIOR test split.

For NWPU VHR-10, the model was fine-tuned through a series of additional training runs, with evaluations conducted at complete-epoch intervals. The experiments covered checkpoints up to 25 overall epochs . Additional fine-tuning experiments were conducted using the epoch-16 checkpoint as the initialization, but did not surpass this result.

For HRSC2016,the model was fine tuned for 25 epochs and the model was evaluated at the complete-epoch checkpoints documented in the corresponding experimental results.

3.4 Oriented Bounding-Box Evaluations

The available oriented results were not all obtained through the same procedure. The DIOR-R result was generated by inference with a GSDet ResNet-50/FPN checkpoint using 900 proposals, two refinement stages, and rotated NMS, and no local fine-tuning was performed. The HRSC2016 contained 444 evaluated images, 1,228 ground-truth objects, and 1,344 detections. The DOTA-v1.0 result file contained VOC-style class-wise AP and COCO-style AP50, AP75, and mAP, but the evaluated split and evaluator configuration could not be established from the file. Because the annotations of the DOTA-v1.0 test images are withheld by the benchmark, so we performed evaluation on test images and then used the DOTA evaluator which in its own site to get the evaluation metrics. We performed only evaluation on DOTA-v1.0 and HRSC-2016.

3.5 Evaluation Metrics and Reporting Convention

COCO-style AP was averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05 unless a source stated otherwise, and AP50 and AP75 denoted AP at IoU 0.50 and 0.75. APs, APm, and APl denoted small-, medium-, and large-object AP under the area definitions of the evaluator. VOC-style mAP was retained under its original label. Percentages were used consistently in the tables, and a dash indicated a metric that was unavailable or undefined. HBB and OBB metrics were not treated as directly comparable, and literature values obtained under different splits or protocols were identified as contextual rather than controlled comparisons.

4 Experimental Results

The experimental outputs were organized by dataset and bounding-box representation. Complete checkpoint sequences were retained where available, and published results were placed beside the local results only when the dataset and detection setting were sufficiently aligned.

4.1 Dataset Analysis

Five aerial-image benchmarks were used or referenced. DIOR and DIOR-R contained 20 categories and 23,463 images, with 9,380 training, 2,345 validation, and 11,738 test images. NWPU VHR-10 contained ten object categories, and 130 test images were confirmed in the local evaluation logs. HRSC2016 was a ship-detection benchmark whose published split was 436 training, 181 validation, and 444 test images, whereas the local HBB logs referred to 453 test images. DOTA-v1.0 contained 15 oriented-object categories and 2,806 original images.

Table 1: Dataset characteristics and image-level split information.

Dataset	Total	Training	Validation	Test	Annotation setting
DIOR	23,463	9,380	2,345	11,738	HBB, 20 classes
DIOR-R	23,463	9,380	2,345	11,738	OBB, 20 classes
NWPU VHR-10	800	520[a]	130[a]	130[a]	HBB, 10 classes
HRSC2016	1,061	436	181	444[b]	Ship detection
DOTA-v1.0	2,806	1,411	458	937	OBB, 15 classes

[a] The NWPU VHR-10 split counts were reproduced from the dataset summary used for this report, and the local manifests were not available for independent verification.
[b] The published split lists 444 test images, whereas the local HRSC2016 HBB evaluation log referred to 453 test images. The published count is shown, and the discrepancy is discussed in the text.
The DOTA-v1.0 values refer to original images, not to patches created by cropping or tiling.

The split information was presented at the original-image level. For NWPU VHR-10, the counts were taken from the dataset summary, although the local manifests were unavailable for verification. For HRSC2016, the published count was retained in the dataset table and the local count was treated as a discrepancy in the experimental records. For DOTA-v1.0, original-image counts were distinguished from the larger number of patches that could be produced during preprocessing.

4.2 DIOR: Horizontal Bounding-Box Detection

The DIOR test-set results recorded for the four complete training epochs are reported in Table 2, and the initial baseline is retained as a separate reference measurement.

Table 2: DiffusionDet results on the DIOR test split. All values are percentages.

Stage	AP	AP50	AP75	APs	APm	APl
Initial baseline	0.317	0.557	0.327	0.2	0.764	0.804
Epoch 1	38.329	58.437	40.807	8.075	31.449	56.314
Epoch 2	41.960	63.063	44.972	9.868	33.178	60.725
Epoch 3	44.560	66.432	47.632	10.500	36.294	63.416
Epoch 4	45.477	67.509	48.829	11.847	36.648	64.234

The aggregate AP increased across the four complete epochs and reached 45.477% at epoch 4, and AP50 and AP75 followed the same trend. The highest APs, APm, and APl among the fine-tuning checkpoints were also observed at epoch 4. The baseline row was preserved as it appeared in the log, and its near-zero values were consistent with a detector that had not yet been adapted to ship detection. The baseline results were retained in their originally reported form and used as a reference for comparison with the experimental results.

Table 3: DIOR HBB comparison with published detectors [8] and the local DiffusionDet result (epoch 4). All values are percentages.

Method	AP50	AP75	APs	APm	APl	AP50:95
Faster R-CNN	73.6	56.5	7.4	30.9	63.0	51.2
Cascade R-CNN	74.1	59.8	7.7	33.6	67.2	54.5
YOLOv3	75.7	60.6	10.1	36.5	67.6	55.5
YOLOv5s	77.7	60.4	11.4	38.4	67.5	55.7
YOLOX-S	76.5	58.2	10.6	37.6	65.3	53.9
YOLOv7-tiny	78.9	62.3	10.1	39.0	69.6	57.6
MSA-YOLO	79.3	63.3	12.6	40.1	70.3	58.6
DiffusionDet (local, epoch 4)	**67.51**	**48.83**	**11.85**	**36.65**	**64.23**	**45.48**

The local DiffusionDet result was lower than all listed published detectors in AP50, AP75, and AP50:95, whereas its APs of 11.85 exceeded those of every listed method except MSA-YOLO. The published detectors were trained for 200 epochs at an input size of 800×800, while the local model was fine-tuned for four complete epochs, so the comparison was interpreted as indicative only.

4.3 NWPU VHR-10: Horizontal Bounding-Box Detection

The complete-epoch DiffusionDet results recorded for NWPU VHR-10 are presented in Table 4 for epochs 1–25. The initial baseline and fractional-epoch measurements were not included.

Table 4: Complete-epoch DiffusionDet results on NWPU VHR-10. All values are percentages.

Epoch	AP	AP50	AP75	APs	APm	APl
Baseline	0.048	0.092	0.034	0.010	0.178	0.013
1	37.806	60.769	41.834	42.538	35.327	34.402
2	46.022	74.330	51.446	43.283	43.126	49.397
3	48.140	78.030	53.135	48.478	44.421	55.031
4	51.330	79.883	57.727	46.307	45.412	63.606
5	54.211	85.385	61.086	47.962	48.103	64.523
6	57.098	86.938	63.927	41.634	50.748	69.037
7	56.893	86.463	64.014	45.750	51.071	67.608
8	59.123	92.439	67.644	52.992	52.281	69.333
9	57.877	88.356	66.930	46.881	51.294	68.691
10	59.066	90.756	67.621	44.847	52.335	69.156
11	58.571	88.384	67.323	41.452	51.832	71.056
12	58.453	88.023	66.373	41.782	50.829	71.059
13	59.488	92.220	65.209	50.768	52.651	68.987
14	58.160	88.130	65.010	41.780	50.440	68.460
15	58.520	88.470	64.350	41.100	50.700	69.510
16	61.760	94.680	69.210	53.160	54.010	71.060
17	58.780	88.880	65.430	46.580	51.490	69.500
18	59.400	90.920	67.570	43.570	51.800	70.050
19	61.410	90.910	69.710	41.940	54.360	72.910
20	61.140	92.890	67.750	39.630	55.370	69.150
21	59.640	88.763	66.511	41.881	52.305	70.494
22	59.005	88.713	64.893	41.634	51.217	71.431
23	60.465	92.823	66.409	42.682	53.326	70.945
24	59.843	88.880	66.658	41.584	53.090	71.194
25	59.628	92.176	64.916	43.926	53.334	69.303

The baseline row was preserved as it appeared in the log, and its near-zero values were consistent with a detector that had not yet been adapted to ship detection. The highest aggregate AP and the highest AP50 were

recorded at epoch 16, at 61.760% and 94.680%, respectively. The highest AP75 and APl were recorded at epoch 19, at 69.710% and 72.910%, while the peak APs and APm were recorded at epoch 16 and epoch 20, respectively. The strongest checkpoint therefore depended on the evaluation metric and the object-size category. Since the epoch was selected from test-set measurements, the epoch-16 result was regarded as the best recorded checkpoint rather than as a validation-selected model.

Table 5: NWPU VHR-10 HBB comparison with published detection methods [9]. All values are percentages.

Method	AP	AP50	AP75	APs	APm	APl
Mask R-CNN	60.6	90.8	69.2	52.2	60.9	54.8
PANet	66.3	91.3	77.3	57.9	57.5	60.2
PointRend	64.9	88.3	76.3	54.0	65.1	60.7
BlendMask	68.0	91.1	79.0	56.7	68.1	57.7
Adaptive/dynamic feature-learning model	69.4	93.1	81.5	58.4	69.5	65.6
DiffusionDet (local, epoch 16)	**61.76**	**94.68**	**69.21**	**53.16**	**54.01**	**71.06**

The published values were reproduced as reported in the source. The local epoch-16 AP of 61.76% exceeded that of Mask R-CNN but remained below those of the other four methods, and the AP75 of 69.21% was comparable to that of Mask R-CNN and lower than those of the remaining methods. In contrast, the local AP50 and APl values were higher than those of all listed methods, and the APs value exceeded that of Mask R-CNN, whereas the APm value was lower than all of them. The comparison was interpreted cautiously because the implementation and training conditions were not matched.

4.4 HRSC2016: Horizontal Bounding-Box Detection

The complete-epoch HRSC2016 horizontal-box results are reported in Table 6 for epochs 5, 10, 15, 20, and 25, together with a separate baseline measurement. The APs value was returned as undefined by the evaluator and is represented by a dash rather than by zero.

Table 6: DiffusionDet HBB results on HRSC2016 at complete epochs. All values are percentages.

Stage	AP	AP50	AP75	APs	APm	APl
Baseline (raw evaluator output)	0.613	1.578	0.417	–	0.198	2.915
Epoch 5	82.3373	93.9734	88.2470	–	51.6265	84.4338
Epoch 10	84.3892	94.9283	90.5782	–	53.8800	86.6134
Epoch 15	83.8449	94.6019	90.2673	–	55.4486	85.8686
Epoch 20	84.1925	94.2290	90.7152	–	56.5653	86.6608
Epoch 25	84.4688	94.4234	90.7618	–	56.4812	86.9604

The baseline row was preserved as it appeared in the log, and its near-zero values were consistent with a detector that had not yet been adapted to ship detection. The strongest aggregate AP was recorded at epoch 25 (84.4688%), the highest AP50 at epoch 10, the highest AP75 and APl at epoch 25, and the highest APm at epoch 20. Since APs was undefined at all checkpoints, no conclusion about small-object performance was drawn.

Table 7: Published HRSC2016 horizontal-box results [6, 11] and the local DiffusionDet evaluation.

Method	Box type	Reported metric / value
SRBBS (NRER-REG-BB)	Horizontal	mAP 55.7
SRBBS (NBEB-REG-BB)	Horizontal	mAP 79.7
SHD-HBB	Horizontal	mAP 69.5
Faster R-CNN	Horizontal	mAP 84.0
DiffusionDet (local, epoch 25)	**Horizontal**	**COCO AP 84.4688; AP50 94.4234**

The published values were reported as mAP, whereas the local metrics were obtained through COCO-style evaluation over multiple IoU thresholds, and the values were therefore not ranked as if measured under one protocol. The split-count discrepancy between the local test records and the published split was retained as a limitation.

4.5 DIOR-R: Oriented Bounding-Box Detection

The DIOR-R local result was generated by inference with a GSDet checkpoint, without local training or fine-tuning. The AP values over ten IoU thresholds are given in Table 8. The evaluator reported a mean of 46.23%, while the mean of the rounded values listed was 46.23%.

Table 8: Local GSDet inference result on DIOR-R across IoU thresholds.

IoU	AP (%)	IoU	AP (%)
0.50	69.70	0.55	67.00
0.60	63.40	0.65	59.40
0.70	54.00	0.75	48.10
0.80	41.00	0.85	31.70
0.90	21.00	0.95	7.10
Mean AP (0.50:0.05:0.95)			46.23

Table 9: DIOR-R oriented-detection comparison reported by GSDet [10], with the local inference result. All values are percentages.

Method	AP50	AP75	mAP	FPS
Oriented RepPoints	66.31	44.36	42.81	29
DCFL	66.41	41.83	40.88	29
H2RBox-v2	57.13	32.13	32.77	30
COBB-sig	65.65	42.78	–	14
ARS-DETR	66.12	45.81	43.89	12
GSDet (900 Gaussians, 1 iteration)	69.35	48.27	46.19	28
GSDet (900 Gaussians, 3 iterations)	70.05	48.91	46.69	21
Local GSDet inference-only	**69.70**	**48.10**	**46.23**	–

The local AP50 and mAP lay between the published one-iteration and three-iteration GSDet values, and the local AP75 was close to the published values. The local run was inference-only on the 11,738 DIOR-R test images, and the exact inference settings could differ from those of the paper. The result was therefore reported as a local baseline evaluation rather than as a reproduced training result or a strict benchmark reproduction.

4.6 HRSC2016: Oriented Bounding-Box Detection

The local HRSC2016 OBB summary contained AP values from IoU 0.50 to 0.95, together with recall, the number of detections, and the number of ground-truth objects. The mean AP was 72.32%. The run summary identified the method as ReDet and recorded checkpoint/state-dict mismatches and a data-prefix issue, and the result was consequently retained as provisional; it was not claimed to be a verified ReDiffDet run.

Table 10: Local provisional HRSC2016 OBB evaluation across IoU thresholds.

IoU	Recall	AP	AP (%)
0.50	0.97638	0.90405	90.405
0.55	0.97476	0.90405	90.405
0.60	0.97068	0.90307	90.307
0.65	0.95928	0.90104	90.104
0.70	0.95358	0.90067	90.067
0.75	0.92752	0.89506	89.506
0.80	0.86971	0.79554	79.554
0.85	0.75814	0.67799	67.799
0.90	0.48453	0.32812	32.812
0.95	0.08795	0.02273	2.273
	Mean AP	0.7232	72.32

The evaluated summary contained 444 images, 1,228 ground-truth objects, and 1,344 detections. The comparative HRSC2016 results reported in the OrientedDiffDet paper [3] were placed beside the local provisional output in Table 11.

Table 11: HRSC2016 oriented-detection comparison with the local provisional evaluation. All values are percentages.

Method	AP	AP50	AP75
DiffusionDet	50.6	76.1	60.6
RoI Transformer	61.4	89.2	75.8
Oriented R-CNN	54.7	89.6	63.7
FCOS	52.9	88.3	62.0
RepPoints	51.8	83.7	59.1
OrientedDiffDet	72.4	90.3	89.5
Local OBB output (using Rediffdet)	**72.32**	**90.405**	**89.506**

The similarity between the local ReDiffDet results and the published OrientedDiffDet results may be attributed to their shared diffusion-based formulation for oriented object detection, in which candidate bounding boxes are progressively refined to improve localization and orientation estimation. Furthermore, the characteristics of HRSC2016, particularly its elongated oriented ship instances, may favour both approaches. However, this numerical similarity does not imply architectural equivalence or establish a validated reproduction of either method.

4.7 DOTA-v1.0: Oriented Bounding-Box Evaluation

The local DOTA-v1.0 output contained class-wise VOC-style AP values for 15 categories and a separate COCO-style summary. The VOC-style mAP was 73.9373%, and the COCO-style summary gave AP50 of 73.9373%, AP75 of 52.3050%, and mAP of 47.6231%. The identical AP50 values indicated that both summaries were computed from the same detections, although the relationship between the evaluator configurations was not fully verified, and the values were kept under their original labels.

Table 12: Local DOTA-v1.0 evaluation metrics retained under the labels in the output file.

Reported evaluator output	Metric	Value (%)
VOC-style task 1	mAP	73.9373
COCO-style	AP50	73.9373
COCO-style	AP75	52.3050
COCO-style	mAP (IoU 0.50:0.95)	47.6231

Table 13: DOTA-v1.0 oriented-detection comparison with published GSDet benchmark methods [10] and the local output. All values are percentages.

Method	AP50	AP75	mAP
Oriented RepPoints	74.38	46.56	44.57
DCFL	73.13	39.62	41.14
H2RBox-v2	72.60	39.67	40.53
COBB-sig	75.52	48.35	45.61
ARS-DETR	74.16	49.41	46.21
GSDet (900 Gaussians, 1 iteration)	75.44	52.15	47.77
GSDet (900 Gaussians, 3 iterations)	75.74	52.40	48.16
Local DOTA output	**73.9373**	**52.3050**	**47.6231**

The DOTA-v1.0 predictions were submitted to the official DOTA evaluation server for evaluation, since the ground-truth annotations of the test set were not publicly available for local evaluation. The reported AP50, AP75, and mAP values were obtained from the server evaluation. Thus, the result corresponds to the official evaluation procedure rather than a locally computed score.

4.8 Summary of Recorded Results

The principal local results and the published references are consolidated in Tables 14 and 15. Horizontal and oriented results were kept in separate rows, and the status of each oriented output was recorded explicitly.

Table 14: Summary of principal local evaluation results. All values are percentages.

Dataset	Task / method	AP or mAP	AP50	AP75	Status
DIOR	HBB, DiffusionDet	45.4766	67.5089	48.8293	Local test evaluation
NWPU VHR-10	HBB, DiffusionDet	61.76	94.68	69.21	Local test evaluation
HRSC2016	HBB, DiffusionDet	84.4688	94.4234	90.7618	Local test evaluation
DIOR-R	OBB,GSDet inference on RediffDet	46.23	69.70	48.10	Inference-only
HRSC2016	OBB, Inference on ReDiffDet	72.32	90.405	89.506	Inference-only
DOTA-v1.0	OBB, Inference on ReDiffDet	47.6231	73.9373	52.3050	Inference-only

Table 15: Local results against the strongest listed published reference per dataset. All values are percentages.

Dataset	Local result	Published reference	Local AP	Ref. AP
DIOR (HBB)	DiffusionDet, epoch 4	MSA-YOLO	45.48	58.6
NWPU VHR-10 (HBB)	DiffusionDet, epoch 16	Adaptive feature learning	61.76	69.4
HRSC2016 (HBB)	DiffusionDet, epoch 25	Faster R-CNN (mAP)	84.47	84.0
DIOR-R (OBB)	GSDet inference on RediffDet	GSDet, 3 iterations	46.23	46.69
HRSC2016 (OBB)	Inference on ReDiffDet	OrientedDiffDet	72.32	72.4
DOTA-v1.0 (OBB)	Inference on ReDiffDet	GSDet, 3 iterations	47.62	48.16

5 Discussion

The horizontal-box experiments showed that DiffusionDet could be adapted to aerial datasets with different object categories and object-size characteristics. On DIOR, the continuation improved the aggregate AP from 38.329% at epoch 1 to 45.477% at epoch 4. On NWPU VHR-10, the highest aggregate AP was 61.76% at epoch 16, and later results fluctuated. On HRSC2016, the highest aggregate AP among the complete-epoch test measurements was recorded at epoch 25. The distinction between validation-based model selection and test-set reporting remained important.

The HRSC2016 OBB evaluation produced AP, AP50, and AP75 values of 72.32, 90.405, and 89.506, respectively. However, checkpoint/state-dictionary compatibility mismatches and uncertainties regarding the `sub_data_root` configuration were recorded during the experiment. Although the evaluation produced numerical results, these issues left uncertainty regarding the checkpoint loading and dataset configuration. Consequently, the results are reported as obtained, with the model attribution retained subject to verification.

The literature comparisons showed that the local DiffusionDet results were competitive on selected metrics but did not exceed all published methods. On DIOR, the local AP was lower than that of every listed detector, including Faster R-CNN, although the APs value was higher than that of all listed detectors except MSA-YOLO. On NWPU VHR-10, the local epoch-16 result showed the highest AP50 and APl among the listed methods, whereas four of the five published methods reported higher overall AP and AP75. For HRSC2016 HBB, the published values were available under different mAP conventions, and a strict ranking was avoided. For DIOR-R and DOTA-v1.0, the local GSDet results were close to the published GSDet values while the differences in test subsets and protocols were preserved.

The following verification issues remained relevant to the final submission. The NWPU VHR-10 training and test split counts could not be confirmed from the available local manifests. For HRSC2016, the local horizontal bounding-box (HBB) evaluation contained 453 test images, whereas the published test split contained 444 images; therefore, the difference in the evaluation subsets required clarification. The HRSC2016 oriented bounding-box (OBB) run also required verification of the checkpoint, state-dictionary compatibility, and `sub_data_root` configuration before its model attribution could be established. The DIOR-R GSDet result was obtained through inference only, without additional local fine-tuning. For DOTA-v1.0, the evaluator configuration and evaluated subset remained to be confirmed. Finally, comparisons between the results reported in the literature and the local experiments were necessarily qualified because the hardware, backbone architectures, training schedules, data augmentations, dataset splits, and evaluation protocols were not identical.

6 Conclusion

Diffusion-based object detection for aerial imagery was studied through horizontal and oriented box formulations. DiffusionDet was fine-tuned and evaluated on DIOR, NWPU VHR-10, and HRSC2016, and the available checkpoint results were tabulated. The best recorded HBB AP values of 45.4766%, 61.76%, and 84.4688% were obtained on the three datasets, respectively. The theoretical designs of ReDiffDet and OrientedDiffDet were examined, and local oriented evaluation outputs were compiled for DIOR-R, HRSC2016, and DOTA-v1.0 on ReDiffDet. We performed only inference on these three datasets for ReDiffDet . We didn't perform any fine tuning for ReDiffDet.The DIOR-R GSDet results were obtained through inference without additional local fine-tuning. The HRSC2016 OBB evaluation was successfully performed using ReDiffDet, and the resulting metrics were retained as the experimental results. For DOTA-v1.0, the predicted detections were submitted to the official evaluation server because the test-set ground-truth annotations were not locally available. The reported metrics were obtained using the server's evaluation procedure.

Further work would focus on strengthening the experimental evaluation and comparability of the results. Additional comparisons could be conducted by rerunning published baselines under identical dataset splits, backbones, input sizes, and training schedules. The evaluation could also be extended through qualitative analysis of predicted bounding boxes, inference-time measurements, GPU memory profiling, and class-wise error analysis.

7 Future Work

Two research directions were prioritized. The first was a *conditional Brownian Bridge diffusion* for oriented detection. Unlike conventional diffusion models, which progressively transform data into Gaussian noise, Brownian Bridge Diffusion Models (BBDMs) construct a stochastic process that gradually transforms samples from a source distribution into a target distribution. The process is conditioned on both endpoints, with the intermediate states following Gaussian distributions whose means interpolate between the source and target and whose variances control the uncertainty during the transition. For oriented detection, the starting distribution could be constructed from image-conditioned candidate boxes, while the target distribution would represent the ground-truth oriented boxes. The main challenge would be designing a stochastic process over oriented geometry that preserves valid box dimensions and rotational properties.

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